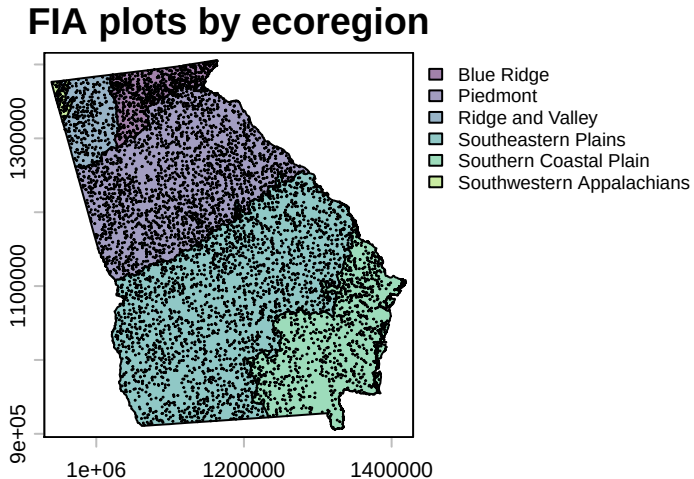


Eyield Demo

Quentin Boccaleri

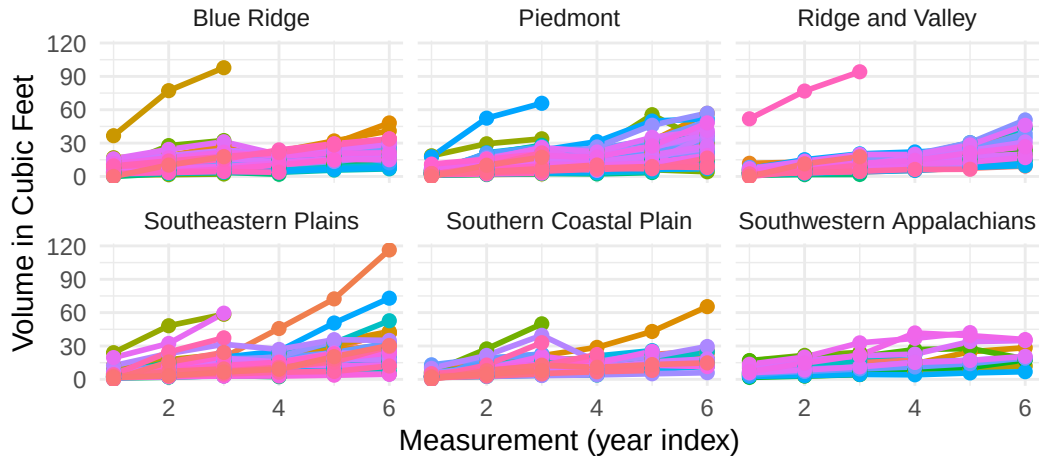
2026-10-01

FIA plots by ecoregion



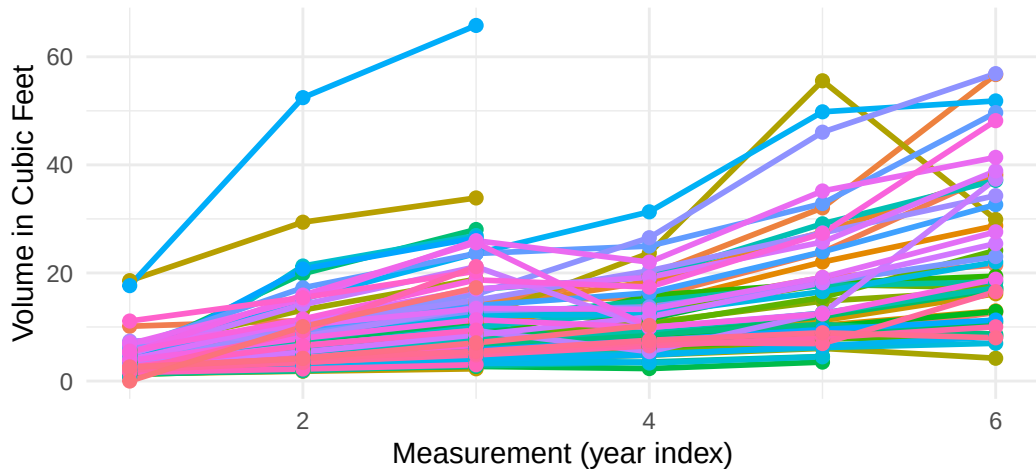
Median growth by species

Median vol by Species



Median growth by species — Piedmont

Median vol by Species – Piedmont



Tree-level equation

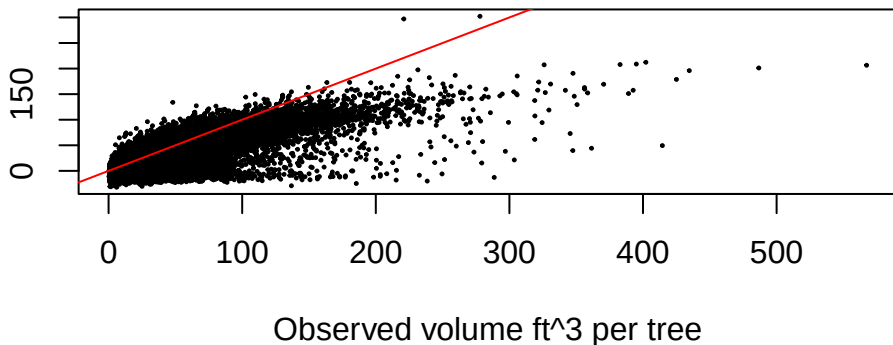
One equation per ecoregion $\times$ consensus cluster:

$$V = b_0 + b_1\text{DBH}_1 + b_2\text{HT}_1 + b_3\text{BA}_1 + b_4t$$

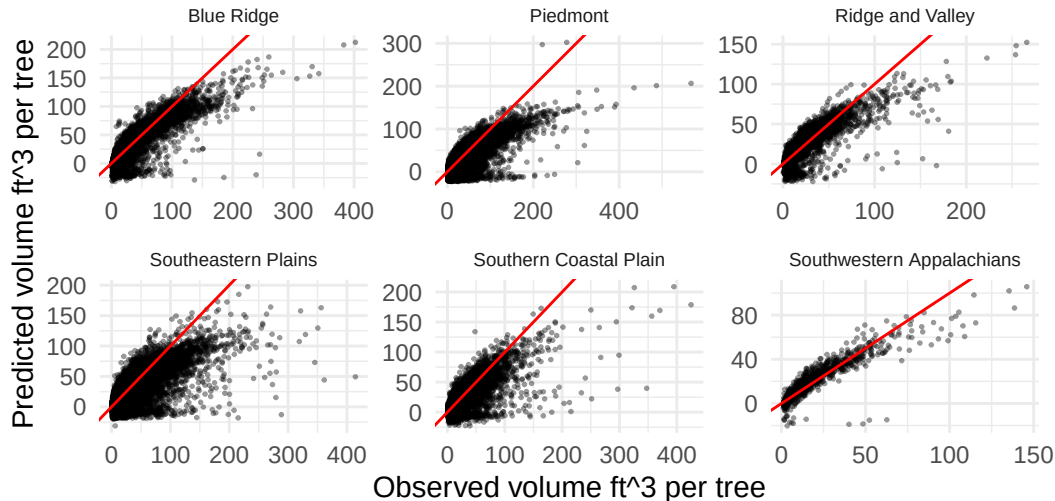
- V = volume (cubic feet) per tree
- DBH_1 , HT_1 = tree DBH and height at the first measurement
- BA_1 = stand basal area per acre at the first measurement
- t = remeasurement periods since the first measurement
- Groups with fewer than 30 observations are skipped

Predicted volume ft³ per tree

Observed vs predicted, all ecoregion x cluster models



Observed vs predicted by ecoregion — tree level



Stand-level equation

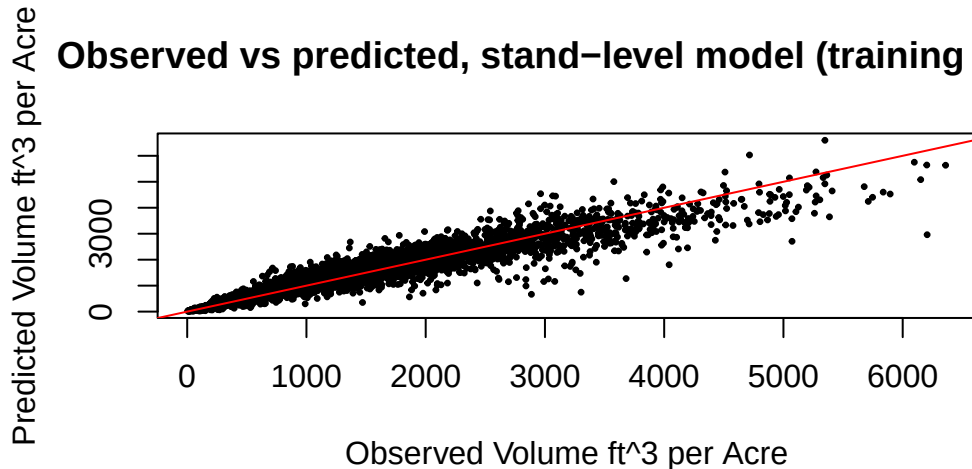
$$\ln(V) = b_0 + b_1 \ln(\text{DBH ratio}_1) + b_2 \ln(\text{HT}_{\max,1}) + b_3 \ln(\text{BA}_1) + b_4 \ln(t)$$

$$\ln(V) = 1.945 + 0.115 \ln(\text{DBH ratio}_1) + 0.337 \ln(\text{HT}_{\max,1}) + 1.011 \ln(\text{BA}_1) + 0.402 \ln(t)$$

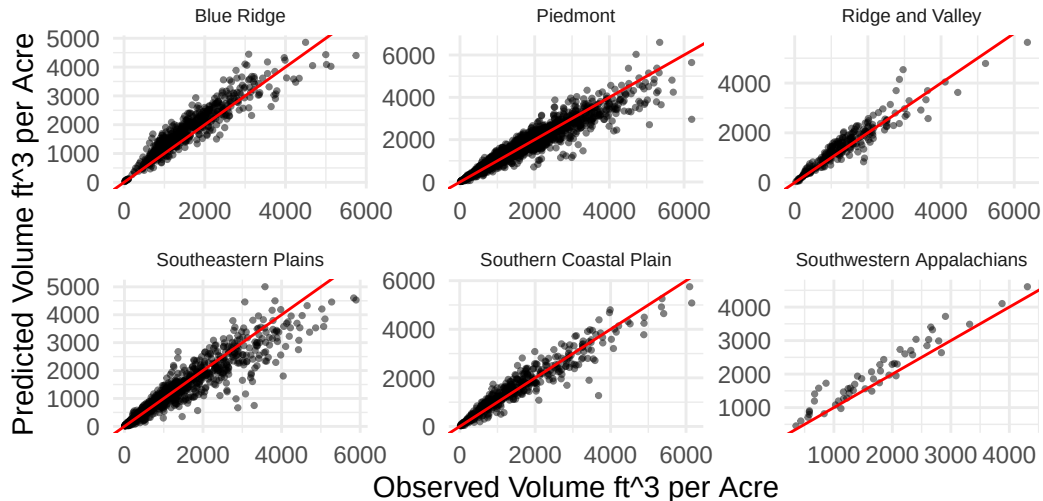
- V = volume per acre; back-transformed with Baskerville correction $e^{\sigma^2/2} = 1.032$
- DBH ratio is the ratio of min. DBH to max. DBH recorded in the first measurement year within the plot
- HT max is the height of the tallest tree in the first measurement year within the plot
- BA is basal area per acre in the first measurement year within the plot
- Adj. R^2 (log scale) = 0.931, $n = 3927$ training plot-measurements

Observed vs predicted — stand level

Observed vs predicted, stand-level model (training plots)



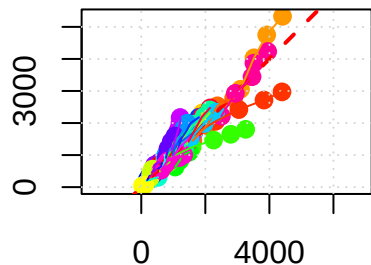
Observed vs predicted by ecoregion — stand level



Stand vs tree-level models — test plots

Predicted Volume ft^3 per Acre

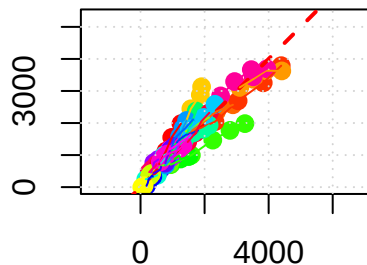
O vs. P – Stand model



Observed Volume ft^3 per Acre

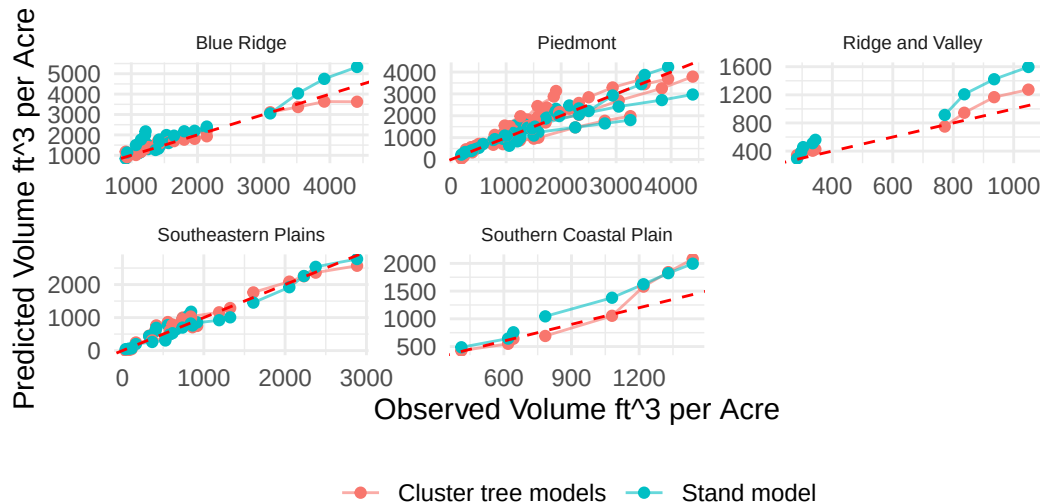
Predicted Volume ft^3 per Acre

O vs. P – Cluster tree mode



Observed Volume ft^3 per Acre

Test plots by ecoregion — stand vs tree-level models



Conclusion

Summary

- Both the stand-level and tree-level models performed decently when applied to novel training data.
- Stand-level has better fit when it comes to training data. -Tree-level had a slightly better RMSE when compared to stand-level for the test data.
- Tree-level models need to be optimized for each cluster to improve the fit.

Next Steps

- Fine tune the hyperparameters for the cluster models.
- Create a system of models (unique optimized model for each cluster).
- Explore alternative forms for the stand-level model.
- Expand across the Southeast.